CRE132 Unity Tutorials

The tutorials and the order to do them in

How the tutorials work

There are 60 tutorials in five sections. Each one teaches one or two ideas in a small Unity scene, and each has its own folder with three files:

- **`CRE132_<Tutorial>_Starter.zip`**: a small Unity project with the scene already set up. This is the one you work in.
- **`CRE132_<Tutorial>_Worksheet.pdf`**: the steps to follow, what each step does, and **Try it** tasks at the end.
- **`CRE132_<Tutorial>_Finished.zip`**: the same project with the tutorial complete, to compare with yours. It comes out after the session.

The two slices at the end of the platformer and top-down sections (P14, T14) have no Starter: you explore and extend the Finished level.

For each tutorial:

1. Unzip the Starter. In Unity Hub, choose **Add > Add project from disk** and pick the unzipped folder. Open it with Unity **6000.3.11f1**.
2. Follow the worksheet from the top: it says which scene to open and what to build.
3. Do the **Try it** tasks. They're where the ideas stick.
4. When the Finished project is out, open it if yours behaves differently, and compare.

The art and sound come from asset packs licensed for CRE132 coursework only (each project's README lists them): don't share them or use them in other projects.

The order

Do the tutorials in the order listed below, top to bottom. Later tutorials use the code and ideas from earlier ones, and some worksheets ask you to reuse a script you wrote in an earlier tutorial. If yours doesn't work, the worksheet says where to find a working copy in that tutorial's Finished project.

1. **Core** (C01–C13): the Unity editor, sprites, physics, animation, and reading the keyboard. C04–C10 read the keyboard the old way (the Input Manager), which keeps the code short; C11 switches to the Input System, used from then on.
2. **Shared systems** (S01–S10): health, menus, sound, saving: code that both kinds of game use, each tested in a small scene with target dummies.
3. **Platformer** (P01–P14): a side-on jumping game, ending with a short complete level.
4. **Top-down** (T01–T14): an action adventure seen from above, ending with a small dungeon.
5. **Effects** (E01–E06): particles, screen shake and other touches that make a game feel good. They use the top-down player, so do them last.

Optional tutorials are extras: you can skip them without missing anything later tutorials need. **Long** ones need more time than usual, so start them early or split them over two sessions. S09 isn't long, but C# events are the hardest idea in the course: take your time over it.

1. Core

Tutorial	What you'll learn	Notes
C01 Sprites and Sorting	Importing sprites, cutting up a sprite sheet, settings for sharp pixel art, which sprite draws in front	
C02 Camera Setup	Camera size, the Pixel Perfect Camera, a Cinemachine camera that follows the player	
C03 Tilemaps	Painting levels with a Tile Palette on several tilemap layers, Rule Tiles	
C04 Input Manager	Reading keys, moving a sprite with `transform` and `Time.deltaTime`	
C05 Physics 2D	Rigidbody 2D, colliders, physics materials, moving the player with physics	
C06 Collisions and Triggers	Collision and trigger messages, layers, tags, a raycast ground check	
C07 Sprite Animation	Animation clips, the Animator Controller, parameters and transitions	
C08 Prefabs	Prefabs, variants and nested prefabs, creating and destroying objects in code	
C09 ScriptableObjects	Data assets that hold stats shared by several objects	
C10 2D Lighting	2D lights, shadows, normal maps	
C11 Input System	The C05 player rewritten with the Input System and an Input Actions asset	
C12 Input to Blend Tree	Driving idle, walk and run animations from the move input	
C13 Skeletal Animation	Rigging a character made of parts with bones	Optional

2. Shared systems

Tutorial	What you'll learn	Notes
S01 Health and Damage	A Health component, damage zones, a dummy that dies at zero health	
S02 Singleton Managers	A GameManager that keeps the score between scenes, and the mistakes to avoid	
S03 UI Basics	Canvas, anchors, TextMeshPro, buttons, a HUD with score and health bar	
S04 Scene Management	Loading scenes, a fade, a loading bar, keeping data between scenes	
S05 Audio	Sound effects, music, an Audio Mixer and a volume slider	
S06 Menus	Main, pause, settings and game over menus	Long
S07 Object Pooling	Reusing bullets and effects instead of creating new ones	Long
S08 Save and Load	PlayerPrefs for settings and high score, a JSON save file	
S09 Events	UnityEvents and C# events, so the UI updates without the player knowing about it	Take your time
S10 UI Toolkit	The S03 HUD rebuilt with UI Toolkit, to compare	Optional

3. Platformer

Tutorial	What you'll learn	Notes
P01 Basic Movement	Running and jumping with physics and the Input System	
P02 Better Jump	Variable jump height, faster falling, coyote time, jump buffering	
P03 Advanced Movement	Double jump and dash	
P03b Wall Jump	Wall slide and wall jump, added to the P03 player	
P04 Platforms	One-way, moving and crumbling platforms	
P05 Ladders and Hazards	Climbing ladders, spikes, falling out of the level	
P06 Player Animation	Idle, run, jump, fall and land animations, flipping the sprite	
P07 Collectables	Coins and gems with a sound, sparkles and score	
P08 Enemy Patrol	An enemy that turns at walls and edges, stomping on it	
P09 Damage and Knockback	Knockback when hit, a short time of invincibility with flashing	
P10 Death and Respawn	Dying, checkpoints, respawning, lives	
P11 Shooting	Firing projectiles left and right, enemies reacting to hits	
P12 Parallax	Background layers that move at different speeds	
P13 Level Complete	A goal, a level complete screen, loading the next level	
P14 Platformer Slice	A short complete level built from everything above	No Starter

4. Top-down

Tutorial	What you'll learn	Notes
T01 Top-Down Movement	Moving in eight directions without gravity	
T02 Directional Animation	A blend tree that faces four directions and remembers the last one	
T03 Y-Sorting	Walking behind and in front of trees	
T04 Melee Attack	A sword swing with a hitbox, Animation Events and a cooldown	
T05 Ranged Weapons	Firing projectiles at the mouse	
T06 Enemy AI	An enemy that idles, wanders and chases (a state machine)	Long
T06b Enemy Attack	Two more states for the T06 enemy: attack and return home	
T07 Pathfinding	The enemy finds its way around walls with a 2D NavMesh	Optional
T08 Damage, Knockback and Death	Knockback, invincibility, enemy drops, player death and game over	
T09 Pickups and Inventory	Hearts, keys and coins in a simple inventory	
T10 Doors and Keys	Locked doors, switches, pressure plates, pushable blocks	Long
T11 NPC Dialogue	Talking to characters in a dialogue box with typewriter text	
T11b Yarn Spinner	The same village with Yarn Spinner: choices, and buying a key	Optional, long
T12 Room Transitions	Moving between rooms and into buildings	
T13 Chests	Opening chests and remembering which ones are open	Long
T14 Top-Down Slice	A small dungeon built from everything above	No Starter

5. Effects

Tutorial	What you'll learn	Notes
E01 Particle Effects	Dust, hit sparks, explosions and sparkles with the Particle System	
E02 Camera Shake	Shaking the camera on hits and explosions	
E03 2D Sprite Shaders	Dissolve, outline and colour tint shaders with Shader Graph	Optional
E04 Hit Feedback	A brief freeze, a white flash and floating damage numbers	
E05 Simple Tweens	Bobbing pickups and popping UI with coroutines and `Lerp`	
E06 Post-Processing	Bloom and vignette	